Data bases 2

Instructors: Sara Comai, Piero Fraternali, Davide Martinenghi

sara.comai@polimi.it

piero.fraternali@polimi.it

davide.martinenghi@polimi.it

Data bases 2, 2026/2027



- Teacher:
  - Sara Comai
    - sara.comai@polimi.it 02-2399-3611
  - Dipartimento di Elettronica Informazione e Bioingegneria (building 20A), 1st floor

- Teaching assistant:
  - Francesco Invernici francesco.invernici@polimi.it

Data bases 2, 2026/2027



Monday 10:15-12:15
Friday 10:15-12:15

- Lectures and exercise sessions may take place on both days

- According to School's recommendations:
  - No streaming / attendance is only in presence
    → streaming on the days of the public transport strike

  - Recording of (possibly old) lessons is provided

Textbooks in Italian

[BD1 + DB2] P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi, R. Torlone

"Basi di dati" (2023)



EBOOK available at VitalSource Bookshelf (https://resolver.vitalsource.com/)

From outside Polimi check how to access it: https://www.biblio.polimi.it/en/services/how-to/access-to-an-electronic-resource

[DB2] P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi, R. Torlone

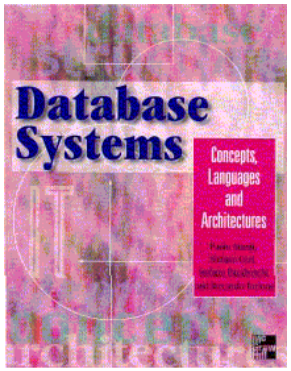
"Basi di dati: Architetture e linee di evoluzione" (2003)

Textbooks in English



[DB1 + DB2] P. Atzeni, S. Ceri, S. Paraboschi, R. Torlone

"Database systems" - McGraw-Hill (1999)



DOWNLOADABLE: http://dbbook.dia.uniroma3.it/

Compared to the last Italian version some chapters are missing

More bibliobraphy on specific topics (e.g., JPA) listed in the official program page of the course

Teaching material



- Material is available on the WeBeep portal

  https://webeep.polimi.it/

  under “Data bases 2”
- Materials include slides of the lectures, exercise sessions, forum, etc.

Prerequisites

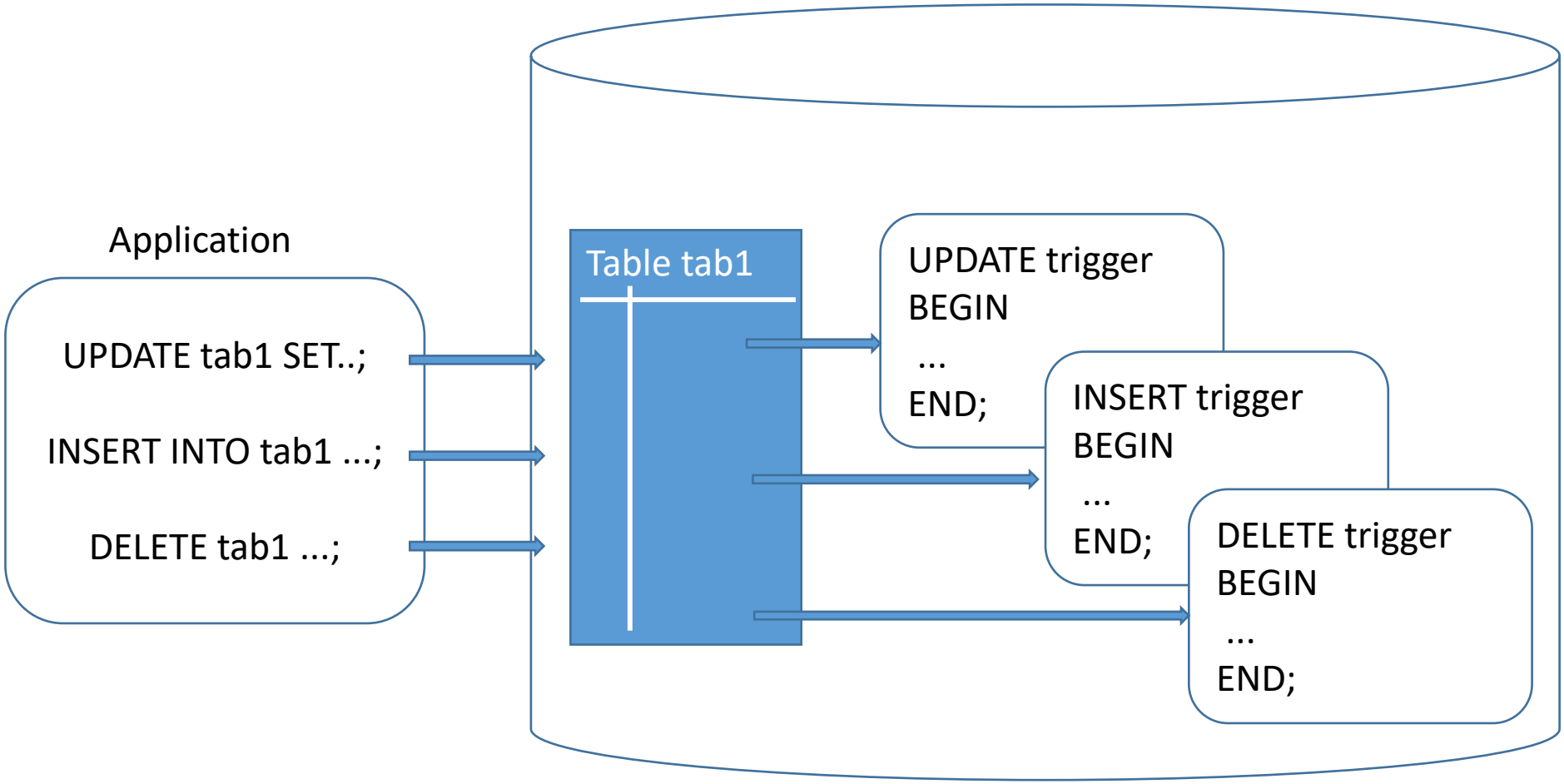


- Basics of Database systems
  - Relational model
  - SQL (SQL-92)
  - Relational algebra

    → See [DB1 + DB2] textbooks

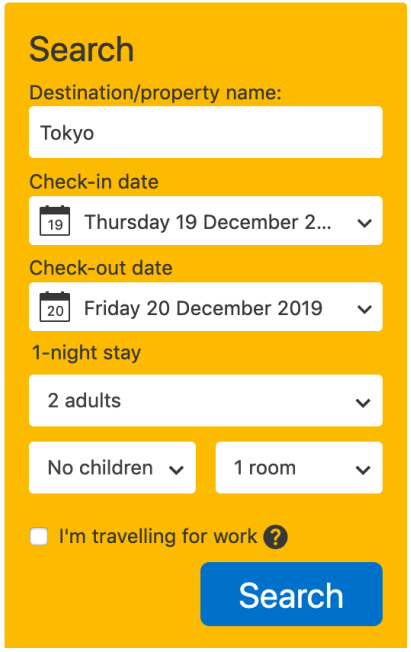
  - Object oriented design with UML and programming with Java (basic)

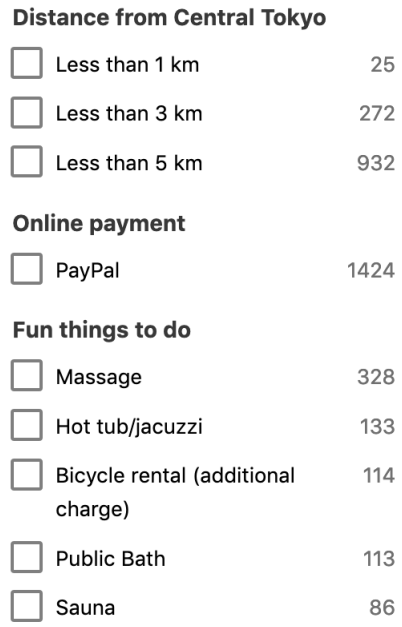
Triggers (SQL)

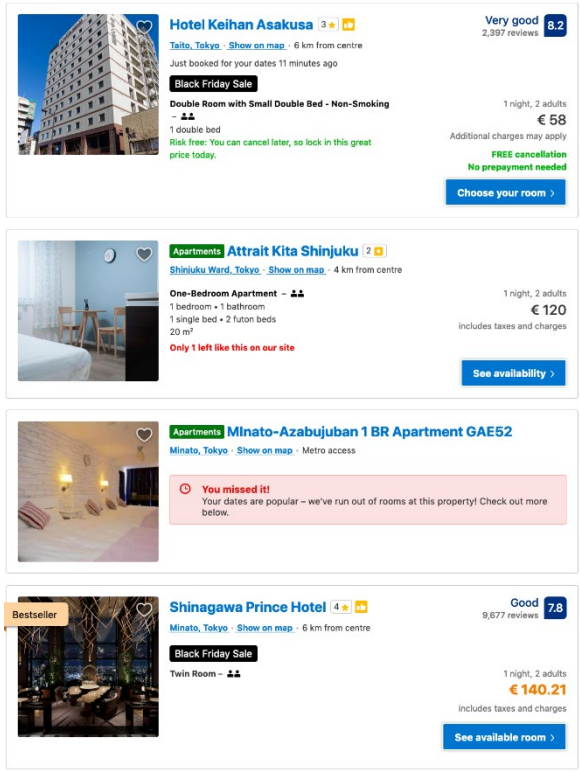


Ranking data

- We are able to store lots of data, but how do we get the *best* data out of a dataset?





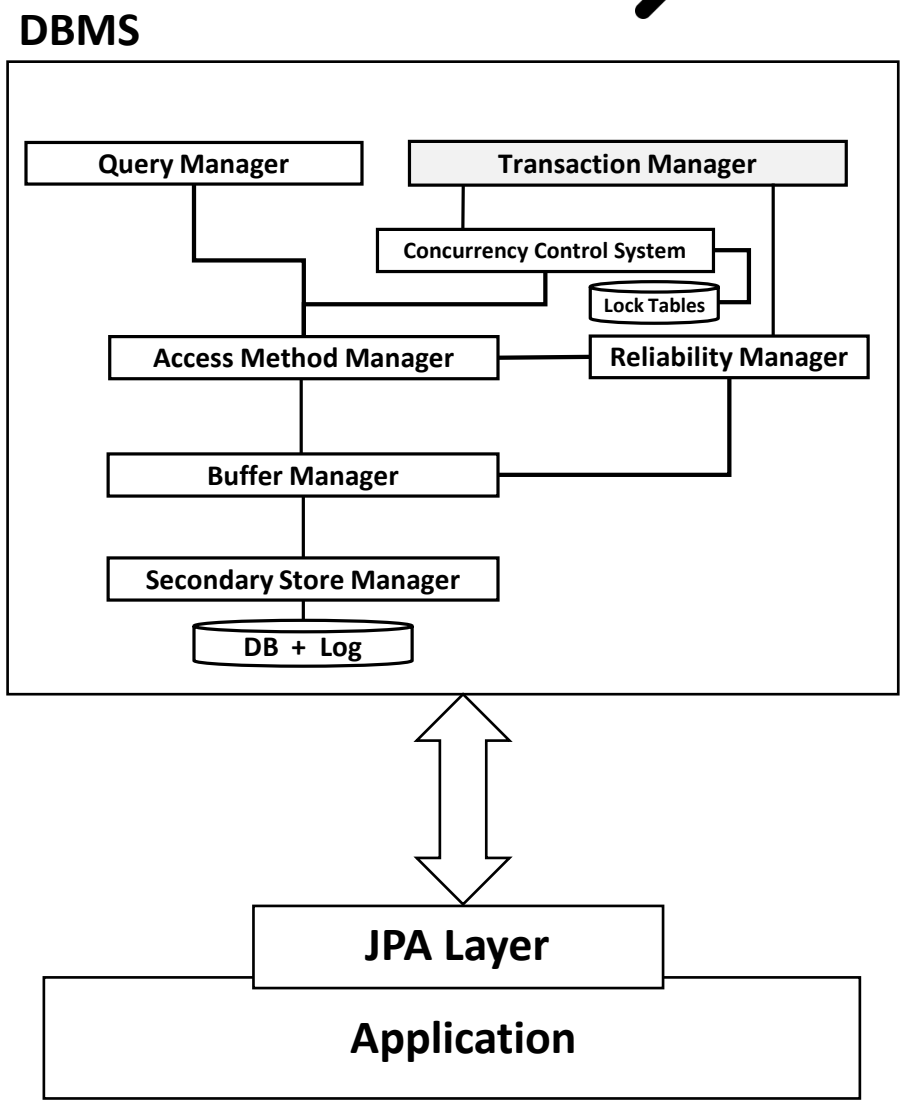


- How does SQL accommodate ranking?

Database architecture and ORM

Opening the database box

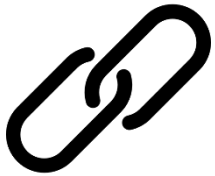
- Physical data management and query optimization
- Transaction management
  - Concurrency control
    - CC theory, locking techniques
  - Reliability control
    - Log and recovery



Opening the application box

- Mapping objects to tables and methods to transactions
- Managing the alignment between object state in main memory and data state in secondary memory
- Building advanced queries

Related topics addressed in other courses



- **056901 - SYSTEMS AND METHODS FOR BIG AND UNSTRUCTURED DATA**
  - Principles and approaches for data storage and management
  - Unstructured data management (NoSQL)
    - graph, semantic, column, document, key-value, IR- based databases
  - Design methodologies for the specification of NoSQL data models

- **056895 - STREAMING DATA ANALYTICS**
  - From data streams and time-series to complex event recognition and processing
  - Streaming Data analytics techniques

Exam



- The exam consists of a **written closed-book** verification covering all the topics of the course (everything published on WeBeep, including materials for flipped classrooms)
  - **3 exercises** on the 5 main topics, possibly with related theoretical questions
    1. Triggers
    2. Ranking and skylines
    3. Physical DBs/Query optimization
    4. DB architecture/technologies (concurrency control etc.)
    5. ORM and JPA

- **No oral exams**
  - Exception: if requested by the teacher

Exam rules



- During the exam:
  - No books, notes, or electronic devices are allowed
  - A calculator is allowed
  - Cheating policies: Communication with other students is strictly prohibited. In this case, students will be asked to exit the classroom (both!).

- In the event of **plagiarism**, students are notified to the appropriate offices and are postponed to a next exam *session*

- Positive marks:

  If a mark is rejected and the written exam is retaken (simply sitting down and seeing the exam text is sufficient), the previous mark is lost!

Exam rules



Please, check the general rules for the exams at Politecnico:

https://www.polimi.it/en/students/didactic-activities-and-degree-examination/exams

Exam – mobility students



- Official rule: *Remote examinations are permitted, at the instructor's discretion and subject to the feasibility of conducting the assessment,* ***only*** *for students* ***officially participating in international mobility programs*** *(documented by a university (Polimi) agreement, such as Erasmus) when the mobility period overlaps with the exam session.*
- Mobility students may take the DB2 exam at the foreign university during their mobility period, at the same time as the exam in Milan, with the **invigilation of an official staff member** (professor or other permanent staff at the university)
- Prerequisite: finding such a local officer and putting me in contact with him/her **at least one week before the exam**
  - No exceptions will be allowed

Tentative agenda

Check WeBeep
for the updated version



Date	Time	#ore	Lec/Exe	Topic	Teacher
14/09/2026	10:15-12:15	2	Lecture	Triggers	SC
18/09/2026	10:15-12:15	2	Lecture	Triggers	SC
21/09/2026	10:15-12:15	2	Lecture	Physical DB	SC
25/09/2026	10:15-12:15	2	Exercises	Triggers	FI
28/09/2026	10:15-12:15	2	Lecture	Physical DB	SC
02/10/2026	10:15-12:15	2	Exercises	Triggers	FI
05/10/2026	10:15-12:15	2	Lecture	Physical DB	SC
09/10/2026	10:15-12:15	2	Lecture	Concurrency Contr	SC
12/10/2026	10:15-12:15	2	Exercises	Physical DB	FI
16/10/2026	10:15-12:15	2	Exercises	Physical DB	FI
19/10/2026	10:15-12:15	2	Lecture	Concurrency Contr	SC
23/10/2026	10:15-12:15	2	Exercises	Concurrency Contr	FI
26/10/2026	10:15-12:15	2	Lecture	Concurrency Contr	SC
30/10/2026	10:15-12:15		Mid-term (other) exams		
02/11/2026	10:15-12:15		Mid-term (other) exams		
06/11/2026	10:15-12:15	2	Lecture	Concurrency Contr	FI
09/11/2026	10:15-12:15	2	Exercises	Concurrency Contr	SC
13/11/2026	10:15-12:15	2	Lecture	Ranking	SC
16/11/2026	10:15-12:15	2	Lecture	Ranking	SC
20/11/2026	10:15-12:15	2	Exercises	Concurrency Contr	FI
23/11/2026	10:15-12:15	2	Exercises	Ranking	FI
27/11/2026	10:15-12:15	2	Lecture	JPA/ORM	SC
30/11/2026	10:15-12:15	2	Lecture	JPA	SC
04/12/2026	10:15-12:15	2	Exercises	Ranking/JPA	FI
07/12/2026	10:15-12:15		Holiday		
11/12/2026	10:15-12:15	2	Exercises	JPA	FI
14/12/2026	10:15-12:15	2	Exercises	Q&A - exercises	SC
18/12/2026	10:15-12:15	2	Lecture	Reliability (flipped)	
21/12/2026	10:15-12:15				